



Monitoring and Understanding Notebook-Based Learning with Jupyter

A two-tier Dashboard for Students and Instructors

Arturo Olivares, Lisa Pawlowski

UNIVERSITÄT
DUISBURG
ESSEN

Offen im Denken

Introduction

What is a Jupyter Notebook?

- **Interactive Computing Environment:** Write and execute code in real-time
- **Supports Multiple Languages:** Primarily *Python*, also *R*, *Julia*, and more
- **Combines Code, Text, and Visuals:** Use *Markdown* for explanations, embed charts, images, and *LaTeX* equations.
- **Ideal for Data Science & Research:** Supports data analysis, visualization, and machine learning workflows.
- Support inquiry-based and interactive learning

Introduction

The Problem: Invisible Learning Processes

- Challenges for instructors:
 - only see final code submission
 - limited insights into how students learn
 - cannot easily identify misconceptions
- Challenges for students:
 - Limited awareness of workflow and error patterns
 - Trial-and-error without reflection
 - Difficulties identifying causes of mistakes

Literature Review

Challenges and Insights from Computational Notebooks

- Computational notebooks widely used in education & research
- Support exploratory coding & flexible workflows, but learning processes are often **hidden** in final submissions
- Systems capturing **cell-level interactions** (execution patterns, time on task, navigation) reveal:
 - Iterative retries & hesitation
 - Fragmented workflows
 - Hidden error patterns
- Novice programmers struggle with error identification & debugging, often relying on trial-and-error.
- Notebook structures may reinforce recurring errors or misunderstandings.

Literature Review

Data collected in previous studies

- Anonymously capture interaction activity:
→ interacting with dashboard: clicking, cell creation/removal, execution data, error, messages, status, timestamp, programming language
- Location within the notebook
- Average time spent on each cell
- Student IDs
- Total number of attempts per student
- Distribution of points per assignment, average effort metric

Literature Review

Visualization

- Real-time data directly within the notebook
 - location of students in the notebook
 - table of contents
 - presenting number of students in cells
- View whole classroom performance:
 - bar charts: number of clicks, executions, error-free executions
 - bubble charts: number of students who have visited cell+ average time spent on each cell
 - cell-level visibility: view activity for specific cell
- Plackback-feature: review students activity for each time-step within a defined time span using a time slider control panel
- Box Plots: distribution of points achieved my students
 - gender filtered to depict variability and central tendency
- Grouped bar charts: for detailed analysis on task level and performance

Methodology

Research Questions

- **RQ1:** Does the inclusion of a learner-facing dashboard in a Jupyter-based programming environment help students better understand and reflect on their own learning process?
- **RQ2:** How do students perceive the usefulness and usability of the dashboard for identifying and correcting mistakes?
- **RQ3:** Which usage indicators can serve as signals for frustration, reflection or effective learning strategies?

Methodology

Two different groups

■ Control Group

No access to Dashboard Extension

No feedback from professor

No help

■ Experimental Group

■ Access to the Extension

→ Monitor their Activity

→ Reflect on their mistakes

→ Adjust Strategy

■ Professor feedback

Methodology

Steps of our study

- I. Initial Survey
- II. First Learning Unit
- III. Second Learning Unit
- IV. Post Survey

Methodology

I. Initial Survey

- 
- Prior Programming Skills
 - Self-confidence on Jupyter
 - Common mistakes they have
 - How are student's usually graded
 - Difficulties
 - What feedback is provided?
 - What content should the dashboard have?
 - Important!
 - No demographic data collected
→ No ethic approval needed

Methodology

II. First Learning Unit

- Basic Python Notebook
 - Basic aspects: Variables, data types...
- Both different groups
- Content of the dashboard? → Depends on the survey
 - Total number of executed cells
 - Error rates
 - Activity timeline
 - ...

Methodology

III. Second Learning Unit

- After feedback, second Python Notebook
 - More advanced: Loops, conditionals, functions...
- Still in two different groups
- *Experimental Group* better than *Control Group*?

Methodology

IV. Post Survey

- Survey needed to assess their experience with the dashboard (not control group)
- Perceived usefulness
- Self-reflection
-

Conclusion

Conclusion

- Not only the final code: much more is known
- Expected
 - Reduced repeated mistakes
 - Better results for the experimental group
 - Teacher better understanding the overall problems

Conclusion

Next Steps

- I. Create and carry out First Initial Survey
- II. Create Dashboard
- III. Recruitment of participants
- IV. Learning units



Monitoring and Understanding Notebook-Based Learning with Jupyter

A two-tier Dashboard for Students and Instructors

Arturo Olivares, Lisa Pawlowski

UNIVERSITÄT
**DUISBURG
ESSEN**

Offen im Denken